JanNaadi

■■■■■ ■■■■■ · People's Voice

Civic Intelligence Platform for Greater Visakhapatnam Municipal Corporation

Hackathon	**Build with AI: Code for Communities — Track 1 (hack2skill)**
City	Visakhapatnam (GVMC) · 98 Wards · 2.1 Million Citizens
Channels	Web · SMS · WhatsApp · Phone Call
Languages	Telugu · Hindi · English · Hinglish · Code-mixed
Status	Live demo via Cloudflare Tunnel · Cloud Run deploy-ready

At a glance — what JanNaadi addresses

What judges look for	JanNaadi's answer
Solves the right problem	5 channels replacing scattered, untracked civic complaints
AI is doing real work	Gemini 2.5 Flash · Cloud STT · Maps Grounding — all live
Can deploy in weeks	One script: ./infra/deploy.sh → Cloud Run, zero servers to manage
Reaches every citizen	Voice + SMS + call — no smartphone or English required
Scales beyond one city	Ward list + plan docs = any Indian city, no code changes

The Problem JanNaadi Solves

Visakhapatnam's 2.1 million citizens have no single place to report civic problems. Complaints land in WhatsApp groups, phone calls, and walk-in visits — never collected, never counted, never connected to what the city has already planned to fix.

Citizen Pain	Ward Officer Pain	City Planner Pain
No channel in Telugu or Hindi	Manual triage with no priority data	Plans disconnected from reality
Feature-phone users fully excluded	Can't tell urgent from routine	Survey data months out of date
Photo evidence goes nowhere	Duplicate complaints not spotted	Investment misaligned with need

JanNaadi's direct response to each pain point

Pain Point	JanNaadi's Answer
No unified channel	5 channels: web text, voice, photo, SMS/WhatsApp, phone call
Language excluded	AI understands Telugu, Hindi, Hinglish, code-mix natively
Feature-phone users left out	Call a number, speak complaint — no internet, no app needed
No priority signal for officers	Ranked dashboard — frequency × severity × recency × ward weight
Plans disconnected from complaints	Discovery Engine links each cluster to Master Plan 2041 clauses

How the AI Works — End to End

Every complaint — regardless of channel or language — passes through this live pipeline. No mock data. No simulations. Tested against real Google APIs.

Step	Google Technology	What It Does for the Citizen's Complaint
1	Cloud Speech-to-Text	Converts voice to text — Telugu, Hindi, English (te-IN / hi-IN / en-IN)
2	Gemini 2.5 Flash	Reads text or photo + caption → outputs category, ward, severity as JSON
3	Google Maps Grounding	'Near RK Beach' → Ward 27 VIP Road (landmark resolved to official ward)
4	text-embedding-004	Turns complaint summary into a 768-dim semantic vector
5	Cloud SQL + pgvector	Groups complaints by meaning — Telugu and Hindi complaints cluster together
6	Discovery Engine	Matches complaint cluster to GVMC Master Plan 2041 — shows relevant clause
7	Cloud Run + Next.js	Ranked dashboard: worst clusters at top, officers see plan clauses alongside

Validation — tested on live APIs, not mocks

8/8	800	192+	<15s
golden text tests pass — live Gemini	synthetic complaints processed end-to-end	semantic clusters auto-formed	per complaint text-to-cluster

• G11: 'near RK Beach' → Ward 27 via Maps Grounding ✓ • G13: Telugu-English code-mix → correct category ✓ • G14: Hinglish complaint → correct ward ✓

• 5 voice clips: Cloud STT → Gemini → cluster ✓ • Twilio SMS + phone call: HMAC-SHA1 verified ✓ • Google OAuth: signin → session → role gate ✓

Built for Every Citizen — Not Just Smartphone Users

5	3+	0	■0
ways to file a complaint	languages te · hi · en · mix	apps to install	data plan needed for phone call

Who	How They Reach JanNaadi	What They Need
City resident	Web — type, voice, or photo	Smartphone + internet
WhatsApp user	Send text, image, or voice note	WhatsApp account
SMS user	Text message to number	Any mobile phone
Feature-phone user	Call the number, speak complaint	No internet, no data, no app
Low-literacy user	Speak in Telugu or Hindi	No reading or typing required
Photo witness	MMS or WhatsApp image + caption	Any phone with a camera

A farmer in Pendurthi with a basic phone calls a number, speaks in Telugu — "nala penchipoyindi" (the drain overflowed). JanNaadi transcribes it, categorises it as drainage in Ward 96, clusters it with similar complaints, and it appears on the officer's dashboard in under 15 seconds. No internet. No app. No English.

Deployed in Weeks — Scales Across India

JanNaadi runs live today on a laptop via Cloudflare Tunnel, with all 5 channels verified end-to-end. Moving to production for GVMC is one command:

```
$ ./infra/deploy.sh
```

• Packages app into a Docker container • Deploys to Google Cloud Run (auto-scale to zero) • Pulls credentials from Google Secret Manager

• Connects to Cloud SQL — managed, backed up • Sets Twilio webhooks to Cloud Run URL • No servers to manage — Google handles everything

Adding a new city — configuration only, no code changes

What Changes per City	What Stays the Same
Ward list loaded into database	AI pipeline — Gemini, STT, Maps, embeddings
Planning documents uploaded to Discovery Engine	Dashboard, clustering, ranking logic
Twilio number registered for that city	Cloud Run hosting, Secret Manager, OAuth
Admin emails set to city officer accounts	Entire codebase — zero changes

Phase	Scope	Timeline
Now	Visakhapatnam pilot — 98 wards, 2.1M people	Live today
6 months	All 13 ULBs in Visakhapatnam district	Config + ward data only
1 year	Major Andhra Pradesh cities	State-level dashboard
2 years	National — 4,000+ urban local bodies	One deploy per city

Impact and What GVMC Needs to Do

2.1M	98	4,000+	192+
citizens in Vizag alone	wards pilot city	urban bodies across India	auto-clusters from 800 test rows

Stakeholder	Before JanNaadi	After JanNaadi
Ward Officer	Manual triage, no priority signal	Ranked dashboard — worst first, auto-sorted
City Planner	Surveys take months, plans disconnected	Real-time clusters + Master Plan clauses
Smartphone user	Scattered WhatsApp, no acknowledgement	Ticket in < 15 seconds, any language
Feature-phone	No digital option at all	Call a number, speak in Telugu
State Govt.	No cross-city intelligence	Federated view across all ULBs

What GVMC needs — JanNaadi does the rest

GVMC Action Required	Already Done by JanNaadi
Register a Twilio phone number	Full pipeline live — all 5 channels verified end-to-end
Approve WhatsApp Business sender (WABA)	Cloud Run deploy script ready — one command
Provide ward officer Google emails	OAuth gate built — @gvmc.gov.in login ready
No other setup needed	8 Master Plan volumes already in Discovery Engine

JanNaadi is ready. Visakhapatnam's citizens are waiting. One phone number, one deploy command, and the city's 2.1 million voices start reaching the right officers — in their own language, from any device.